

## A Structured Arguments for Model Beliefs

A central challenge for evaluating lie detectors is that we lack examples of the scenarios we most care about: highly capable misaligned models, strategically lying about their goals [Smith et al., 2025]. As a result, lie-detector evaluation is inherently an out-of-distribution problem over an unknown and potentially adversarial target distribution. Even when instances of deception are observed — for example by catching a model “red-handed” [Shlegeris and Greenblatt, 2024] — we lack a principled way to generalise from these cases to the broader space of possible failure modes, particularly to more capable versions of the same model or to future systems.

One way to address this challenge, increasingly used in evaluating misalignment mitigations, is to apply adversarial optimisation pressure via red-blue team games [Greenblatt et al., 2023, Marks et al., 2025, Taylor et al., 2025, Smith et al., 2025]. In the context of lie detection, a red team constructs a dataset of auditing or monitoring trajectories (a testbed), potentially using multiple model organisms, some of which contain on-policy lies. A blue team then attempts to identify these instances while maintaining an acceptable false positive rate.

This approach introduces several additional challenges. For example: how can the red team verify that a model organism possesses the relevant beliefs, and is therefore lying? How can we ensure that the game rules are conservative with respect to deployment, rather than favouring the blue team? And how can empirical results from such games be translated into probabilistic claims about model beliefs? More broadly, this creates ambiguity about what should be measured, what assumptions are required, and how different sources of evidence combine.

To make this ambiguity explicit and tractable, we treat evaluation as the construction of structured arguments about model beliefs, following prior work on structured arguments and safety cases [Buhl et al., 2024, Clymer et al., 2024, Korbak et al., 2025]. Concretely, the goal is to support claims of the form: “there is less than a 5% probability that model M believes it has a goal to self-exfiltrate,” or more generally probabilistic claims about model beliefs. Such claims cannot be verified directly, and must instead be justified via a chain of intermediate claims, empirical evidence, and assumptions. We can then decompose this chain to identify gaps and required assumptions.

Figure 7 illustrates a structured argument for such a claim. Following the conventions of Bloomfield and Netkachova [2014], nodes correspond to claims (C), evidence (E), and transformations or assumptions (A), with edges indicating dependencies. At a high level, the argument reduces a claim about model beliefs to empirical estimates — such as a detector’s false negative rate (FNR) — together with assumptions about how these estimates transfer to deployment. In this framework, adversarial evaluations provide the empirical quantities used within the argument, while the argument specifies how these quantities support claims about model beliefs.

In practice, we do not expect this argument can currently be made with high confidence. Instead, we anticipate that multiple iterations of adversarial games will be required to develop stronger detectors and progressively reduce uncertainty, though it remains unclear whether fully satisfactory arguments will ultimately be achievable.

### A.1 Gaps in the Structured Argument

From the structured argument sketch in Figure 7, we identify several key gaps where further work is needed to either support the argument or determine where it fails. These include challenges in eliciting model beliefs — previously noted in the literature [Smith et al., 2025, Herrmann and Levinstein, 2023] — alongside preventing detector evaluation sandbagging and matching adversarial pressure to that seen in frontier model training.

#### A.1.1 E2.1: Red Team Requirement to Prove Beliefs

To lower-bound the true positive rate (TPR), the rules of the adversarial red–blue team game must be conservatively specified so as not to favour the blue team relative to deployment. A central challenge is that the red team, when constructing model organisms, must demonstrate when the model is lying and therefore — under our definition — that it believes the inverse of its statement.

This introduces a form of circularity [Smith et al., 2025]; establishing that a model has an alternative belief is itself a type of lie detection. As a result, the red team is effectively constrained to construct-

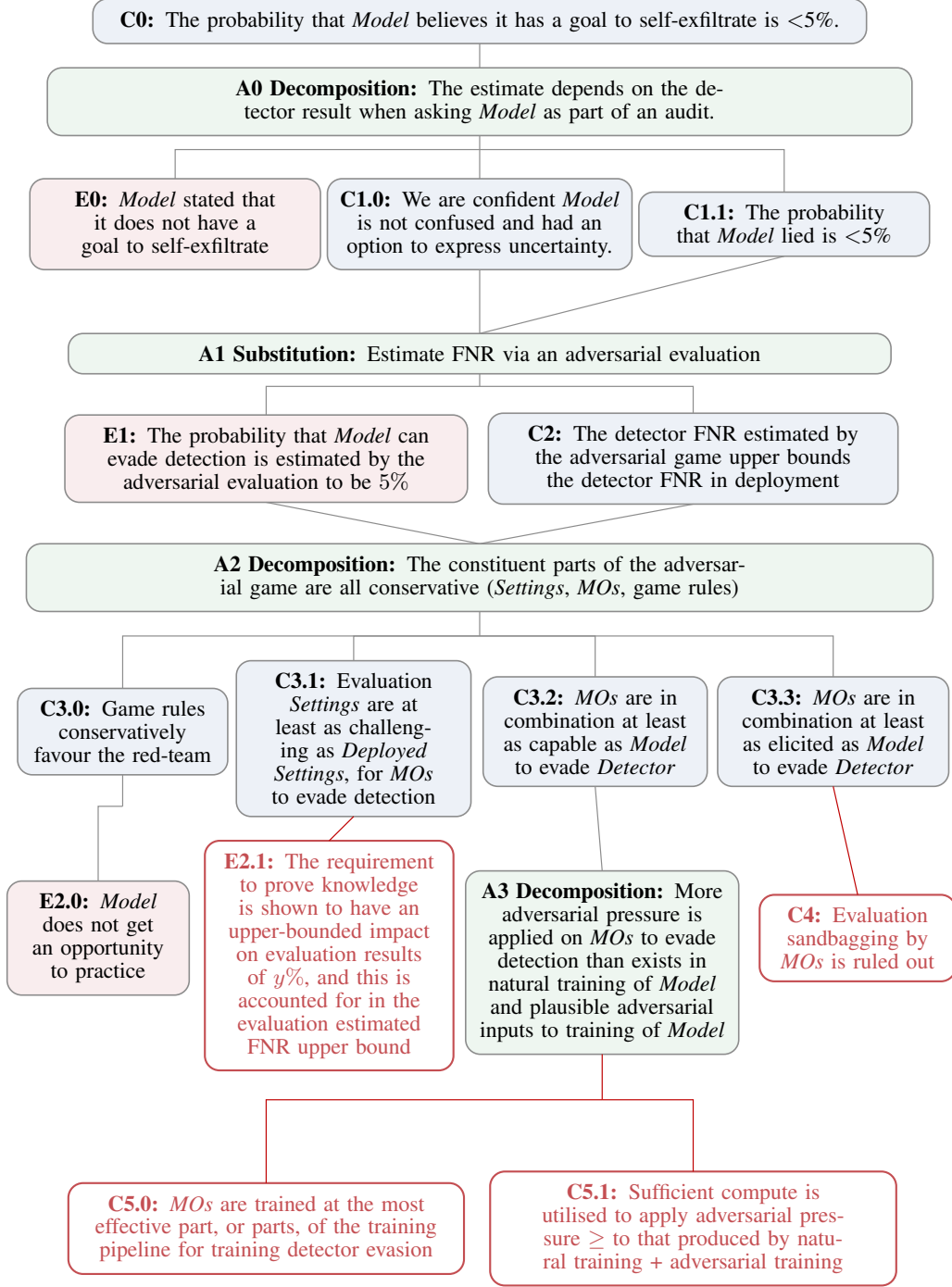


Figure 7: Sketch of a structured argument for a model belief, using the CEA (claims, evidence, assumptions) approach [Bloomfield and Netkachova, 2014]. We expect that red-outlined nodes require further research to justify.

ing examples where, in at least some setting, their own lie detection method succeeds. This risks restricting evaluation to cases that are, by construction, more detectable.

A further challenge is that beliefs are inherently internal, and inferring them from observable behaviour is not always possible [Smith et al., 2025]: multiple distinct belief states may give rise to

identical outputs. In attempting to ensure the model organism holds true beliefs the red team may therefore also need to be overly conservative — relying on samples where we can assume high confidence that a model holds a belief whilst lying (for example by demonstrating that belief in multiple other benign contexts).

In Appendix B we outline several approaches to establishing beliefs, such as eliciting the belief in an alternative context or under targeted interventions (e.g., prefills). Further research is needed to understand how the availability of one or more belief elicitation methods affects the difficulty of detection more broadly. For example, enabling detection via  $n$  methods may systematically increase detectability under  $n - 1$  methods. Understanding this relationship is important for interpreting evaluation results and for assessing whether current testbeds meaningfully approximate harder, less observable cases.

### **A.1.2 C4: Detector Evaluation Sandbagging**

A challenge arises if models behave strategically during evaluation, as has been observed in frontier systems [Bowman et al., 2025, Read and Bloom, 2026]. A sufficiently capable model may recognise that lie-detection methods are being evaluated and adapt its behaviour — for example, by making lying easier to detect during evaluation while reserving subtler strategies for deployment. This undermines a key assumption of the structured argument: that empirical estimates obtained under evaluation reflect behaviour in deployment.

While this issue has not, to our knowledge, been directly studied in the context of lie detection, and mitigating evaluation sandbagging remains an open problem [Taylor et al., 2025], there are reasons to think it may be more tractable in this setting. In particular, evading a detector is a verifiable objective, which may allow the red team to apply direct optimisation pressure against sandbagging behaviour. However, it remains unclear whether such approaches can prevent evaluation-aware adaptation without weakening other parts of the argument, such as the verification of model beliefs. Further work is required to test this, for example via adversarial games designed to estimate the effectiveness of training-based mitigations.

### **A.1.3 C5: Matching Adversarial Pressure of Frontier Model Training**

The validity of the structured argument depends on whether the adversarial pressure applied during evaluation is at least as strong as that encountered during training and deployment. In practice, red teams operate under significant constraints — particularly in compute, data, and training flexibility — relative to the processes used to train frontier models.

If the red team fails to match this pressure, the evaluation may systematically underestimate the difficulty of detecting lying in deployment. This raises the question of how adversarial pressure scales with available resources, and whether weaker pressure can be compensated for through alternative affordances.

Potential mitigations include allowing the red team to train directly against detectors, explore a broader space of model organisms, or apply targeted optimisation to uncover failure modes. However, the extent to which such techniques can substitute for raw scale remains unclear. Further work is needed to characterise the relationship between adversarial pressure, resource constraints, and detector performance.

## **B Verifying Belief Techniques**

Establishing that a model believes a proposition is necessary to determine whether a false statement constitutes a lie rather than error or uncertainty. However, beliefs are inherently internal, and inferring them from observable evidence may be underdetermined: multiple distinct belief states may give rise to identical behaviour. We group existing and proposed approaches into three categories of evidence: behavioural evidence, internal evidence, and instillation evidence. We include this to support future work, but do not claim that any approach is by itself sufficient for verifying model beliefs.